Chapter 8
Measurement Uncertainty in Conformity Assessment



8.1 Statement of Conformity and Decision Rules

Statements of conformity and decision rules are important issues to discuss, especially after the last edition of ISO/IEC 17025: 2017, which made the requirements more rigorous for these issues.

Based on document ILAC-G8:09/2019—Guidelines on Decision Rules and Statements of Conformity, we can establish the following definitions:

Statement of conformity: is an expression that describes the state of conformity or non-conformity with a specification, standard, or requirement.

Decision rule*: a rule that describes how measurement uncertainty will be accounted for when stating conformity with a specified requirement*. (ISO/IEC 17025:2017 clause 3.7)

As we can see, we need to adopt a decision rule to declare a product, process, or measurement standard "conforming."

The ISO/IEC 17025 standard was first published in 1999. Since then, statements of conformity based on specifications or standards have become increasingly required, following the evolution of documentation on the concept of rule decisions used for such statements.

The current standard cites various requirements for the statement of conformity; however, no "unique" rule can address all conformity statements throughout the scope of testing or calibration.

Additionally, professionals (auditors and audited) have doubts about understanding and writing. How do we obey, that is, what laboratories need to apply to meet the customer's requirements and the ISO/IEC 17025 standard?

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8.2 Conformity Assessment

Conformity assessment[1] is performed in tests, inspections, and calibrations to ensure the compliance of products, materials, services, and systems regarding requirements defined by standards, regulations, and legal frameworks, being adopted to establish confidence for consumers and safety and quality of life.

Evaluating conformity is essential in the global economy, because it implies accepting or rejecting items that impact risk analysis, business decisions, financial costs, and sometimes reputation costs.

The result of a measurement can be used to decide whether a variable of interest complies with a specific requirement, and this variable may be, for example:

(a) **The value of a standard 200 g mass of class E2.**

According to the OIML R 111-1 Edition 2004 recommendation, a standard mass of 200 g of class E2 should have a maximum permissible error of 0.3 mg. This is to say that the mass in question must have a value between [199.9997 to 200.0003] g.

(b) **The indication error of a digital voltmeter.**

The value indicated in the standard 999 V.
The value indicated in the voltmeter 1000 V.
Error = 1 V.

(c) **The pH value of a solution.**

Example: pH = 7.474.

These variables of interest have values usually within tolerance limits, called tolerance intervals. If the variable's actual value is within the tolerance range, it is considered "conforming"; otherwise, it is considered "non-conforming."

In practical situations, to perform conformity assessment (e.g., conformity with geometric tolerances), are necessary objective criteria called "decision rules" (which consider a probability of occurrence), which define a "conformity zone" and an "acceptance zone" (which are the results plus measurement uncertainty).

A traditional approach to a decision rule involves comparing a single limit (or limit interval) with the result of a single measurement.

Currently, the probabilistic approach to measurement, which introduces uncertainty as a parameter that expresses measurement variability, significantly affects the decision-making process. See Fig. 8.1 for example.

Figure 8.1 presents four possible measurement results and their uncertainties within a tolerance interval to which we must apply a decision rule. We can undoubtedly say that case (a) is "conforming" and case (d) is "non-conforming"; however, in

[1] According to ISO/IEC 17000: 2004, conformity assessment is any activity performed to determine, directly or indirectly, whether a product, process, system, person, or body meets relevant standards and complies with the specified requirements.

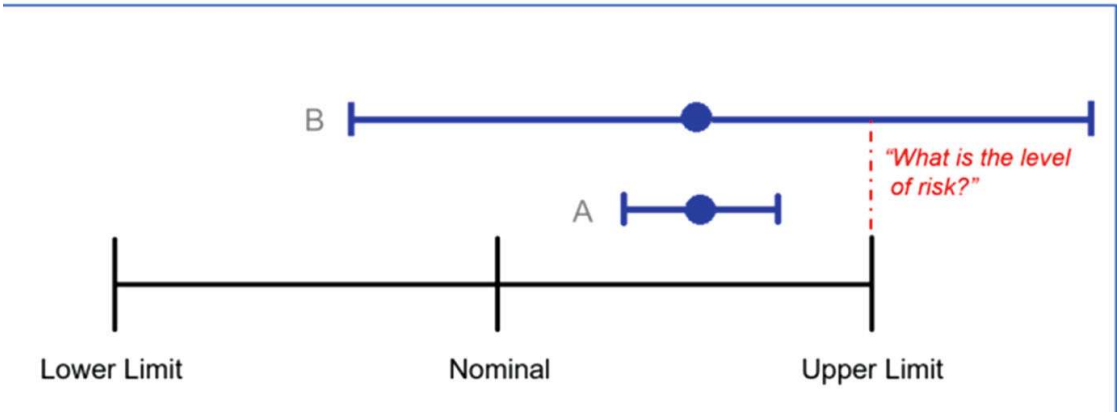


Fig. 8.2 Measurement decision risk. (Source: ILAC-G8:09/2019)

8.3 Uncertainty of Measurement and Risk of Decision

When performing a measurement, there are two situations when stating conformity: inside or outside tolerance regarding the manufacturer's specifications, approved or disapproved in a specific requirement.

In real life, as much as the instrument is used, however accurate and precise, we know that all measured value has an associated measurement uncertainty (U).

Figure 8.2 shows two measurements of the same measurand, but with different measurement uncertainties (the central point is the measurement value, and the horizontal bar shows the measurement uncertainty).

Note that the measurement result (measurement value + measurement uncertainty) in case A is entirely within the tolerance limit. In case B, which has significantly greater measurement uncertainty, the risk of accepting a false result exists. What is the risk level of a false result?

Further, we will discuss the types of errors, risks, and possible decision rules in detail, but now we will anticipate some considerations.

A binary decision rule exists when the result is limited to two options (approved/disapproved, right/wrong, passes/does not pass), and a non-binary decision rule exists when, of course, we have several alternatives to express the result (approved, disapproved, conditional approval, conditional failure).

Analyzing Fig. 8.1, where measurements (a) and (b) are considered approved, and the values (c) and (d) failed in a binary statement with a simple acceptance rule, statements of conformity may be reported as:

- *Pass* (a) and (b): The measured value (central point) is within the specification limits.
- *Fail* (c) and (d): The measured value is outside the specification limits.

In this case, the expanded uncertainty of measurement U was ignored, and the decision was made only based on the measurement value.

Note that even cases (a) and (b), considered approved, have a percentage risk of providing values outside tolerance limits. This percentage is much lower in case

(a) than in case (b), but it exists. Remember that measurement uncertainty is calculated for a probability of 95.45 % and that we have a 4.55 % probability that the declared value is outside the measurement interval.

Thus, it is necessary to study the decision errors assumed when accepting or rejecting a measurement result against specification limits (tolerance intervals). Next, we will analyze these mistakes.

8.4 False Positive and False Negative

When performing a conformity assessment, there are probabilities related to two types of incorrect decisions:

(a) Accept an incorrect result—*false positive.*
(b) Reject a correct result—*false negative.*

Figure 8.3 has four measurement results with their respective uncertainties.

Accepting or rejecting an item when the measured value of your property is close to the tolerance limit may result in an incorrect decision and undesirable consequences.

Analyzing situation (b) in Fig. 8.3, we see that the measured value is below the tolerance limit, but the true value is above, configuring a false positive situation. That is, a situation where we believe the product is within the specification, but it is not. The situation (c) is opposite. We have the value measured above the tolerance limit but below the true value. In this case, we have a false negative situation. That is, we believe that the measurement does not meet the specifications but meets.

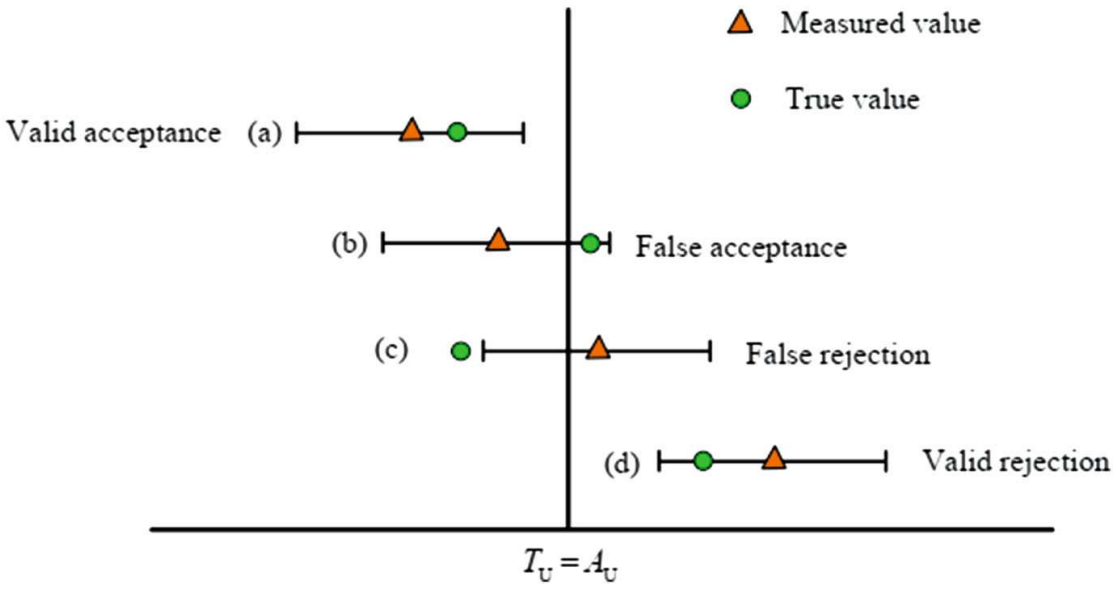


Fig. 8.3 T_U is the upper tolerance limit, and A_U is the upper acceptance limit. There were four measurement results, with their respective uncertainties at 95%. (Source: JCGM 106:2012)

We have consumer risk in the case of a false positive (type II error or β error). In the case of a false negative, we already have producer/supplier risk (type I error or α error).

For measurements, the probability of not accepting an item as shown in Fig. 8.3, (b) or rejecting an item as shown in Fig. 8.3, (c) can reach 50 %. This would happen, for example, if the value measurement of a property were very close to the tolerance limit. In this case, about 50 % of the probability of the measurement result would be on both sides of the tolerance limit, so whether the item was accepted or rejected, there would be a 50 % chance of an incorrect decision.

Any of these odds can be reduced, but at the cost of increasing the other, choosing acceptance limits, which removes the result from tolerance limits.

This is a compliance decision strategy called Guard Band.

8.5 Decision Rules and Guard Band

When measurement is very close to tolerance limits or when uncertainty is significant, an acceptance criterion only considering the upper and lower tolerance limits (simple acceptance) can lead to a high risk of an incorrect decision. Often, more confidence is necessary in accepting or rejecting an analyzed item. For these situations, we can use a guard band.

The guard band is a protection created to remove (reject) values near tolerance limits. Acceptance and rejection zones can be determined, as shown in Fig. 8.4.

(a) **High confidence in correct rejection.**

Only measurements that exceed the guard band placed after the tolerance limit will be rejected in this case.

(b) **High confidence in correct acceptance.**

Only measurements inferior to the guard band placed before the tolerance limit will be accepted in this case.

The region between the upper tolerance limit and the higher acceptance limit is called the *guard band*, reducing the risk of an incorrect decision.

The use of guard bands provides a straightforward way to define decision rules; choosing the size of the guard band defines an acceptance zone that can be used for decision-making. The guard band is generally defined as the expanded uncertainty of measurement (U). It can also be described as zero. This is called simple acceptance or "shared risk."

Figure 8.5 shows the acceptance and rejection zones. The guard band has been chosen so that, for a sample that is in accordance, there is a high probability that the measurement is within the specification limits; this is high confidence in correct acceptance.

Figure 8.5 shows the relative positions of specification limits and acceptance and rejection zones for high acceptance confidence.

The decision to accept an item as in accordance or reject it as not as, according to the specification, is based on a measured value of a property of the item about a declared decision rule that specifies the role of measurement uncertainty in the formulation of acceptance criteria. An interval of measured property values that results in acceptance of the item is called the acceptance zone (see Fig. 8.5), defined by one or two acceptance limits (see Fig. 8.4).

The limits of acceptance and the corresponding decision rules are chosen to manage the undesirable consequences of incorrect decisions. Several widely used decision rules are simple to implement. They can be applied when knowledge of a property of interest is summarized in terms of a better estimate and corresponding coverage interval. Two of these decision rules are described below.

8.5.1 Decision Ruler Based on Simple Acceptance

An important and widely used decision rule is simple acceptance or shared risk. In this rule, the producer and user (consumer) agree, implicitly or explicitly, to accept as (and reject otherwise) an item whose property has a value measured within the tolerance range. With the alternative name of "shared risk," the producer and the user share the consequences of incorrect decisions.

In practice, to maintain the chances of incorrect decisions at acceptable levels for both the producer and the user, there is usually a requirement that measurement uncertainty is considered sufficient for the intended purpose.

One approach to such consideration is to require, given an estimated measured quantity, which expanded uncertainty U for a probability of scope, for example, of 95.45 %, must satisfy:

$$U \leq U_{\max}$$

Where $U_{\max}$ is a mutually agreed expanded uncertainty, this approach is explained by the following situation.

Legal Metrology's decision rule based on simple acceptance has been used to verify instrument measurement.

Consider an instrument that must have an indication error in the interval $[-E_{\max}; +E_{\max}]$. The instrument is accepted under the specified requirement, if it meets the following criteria:

(a) Analyzing the measuring instrument calibration certificate, its measurement error E will be accepted if it satisfies the condition: $E \leq E_{max}$.
(b) Expanded uncertainty will be accepted if it is less than 1/3 of the maximum error.

$$U \leq U_{max} = E_{max}/3$$

8.5.2 Decision Rules Based on Guard Bands

The difference between a tolerance limit (T_U) and a corresponding acceptance limit (A_U) defines the width of a guard band (w).

$$w = T_U - A_U. \tag{8.1}$$

When we consider that a decision rule is protected, that is, it has a small probability of generating false positives or false negatives, we adopt $w > 0$. (Fig. 8.6).

8.6 Unilateral Tolerance Interval with the Normal Curve

The probability of a measurement (product, component, etc.) being in compliance depends on the knowledge of the measuring X and its respective probability density function (PDF) $P(X)$. In most cases, it is reasonable to characterize *the* knowledge X by a normal distribution, and thus, we can calculate its probability.

If the production distribution is normal and a normal distribution also characterizes the measurement system, then the distribution function $P(X)$ will also be normal.

More generally, if the probability function is characterized by a normal distribution and the previous information is insufficient, then the posterior PDF (post-measured) will be approximately normal. In this case, $P(X)$ may be adequately approached by a normal distribution with the mean $\bar{x}$ and the standard deviation by standard uncertainty $u(x)$, calculated according to ISO Gum criteria.

Assuming, then, that the PDF $P(X)$ for the measurement X is (i.e., well approximated by) a normal distribution specified by a mean $\bar{x}$ and a standard uncertainty $u(x)$, we will have:

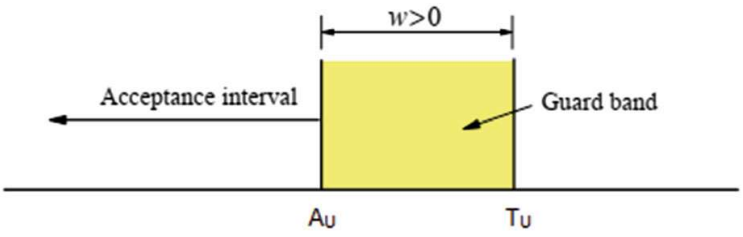


Fig. 8.6 Guard band-based decision rule. An upper A_U acceptance limit within an upper T_U tolerance limit defines an acceptance range that reduces the probability of false acceptance of a non-conforming item (consumer risk). By convention, the length parameter w associated with a guard band is considered positive: $w = T_U - A_U > 0$. (Source: JGCM 106:2012)

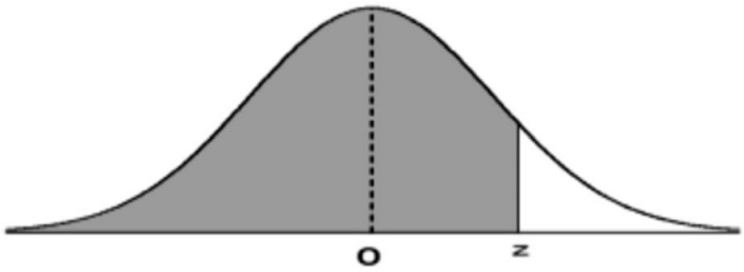


Fig. 8.7 Standard normal accumulated probability

$$p(x) = \frac{1}{u\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-\bar{x}}{u}\right)^2} = \varphi\left(x; \bar{x}; u^2\right) \tag{8.2}$$

The probability of X being in the interval [a, b], with a normal PDF $P(X)$, will be:

$$\mathrm{P}(a \le X \le b) = \varphi\left(\frac{b-\bar{x}}{u}\right) - \varphi\left(\frac{a-\bar{x}}{u}\right) \tag{8.3}$$

The probability $P(X)$ can be found in a standard normal accumulated table or through MS EXCEL© software, function DIST.NORM.N (x, mean, standard deviation, cumulative).

Figure 8.7 and Table 8.1 show the standard normal accumulated probability of z from 0.00 to 3.99.

Next, we will apply Eq. (8.3) to estimate the probability of obtaining a value accepted as valid for a given specification from a measurement result.

8.6.1 Examples of Probability Estimation in Simple Acceptance

(a) **Single upper tolerance limit—T_U.**

As we saw at the beginning of this chapter, a measurement process's tolerance is the maximum variation admitted by the process variables. This tolerance range is the limit within which the parameters of interest must be located.

In some situations, we do not work with a tolerance range but with a single tolerance value. Given a single upper tolerance limit (T_U) and an estimated y measurement with standard measurement uncertainty $u(y)$, a decision rule must define a probability of compliance (Pc) assuming a false negative (products in compliance are incorrectly rejected).

The expression to be tested is:

$$Pc = P(y \le T_U) = \varphi\left(\frac{T_U - y}{u}\right) \tag{8.4}$$

The question is: Is it possible to determine minimum and maximum values within the tolerance range, where all values found will comply with the criterion adopted?

To answer this question, the concept of a guard band was created.

8.6.2 Examples of Probability Estimate Using Guard Bands

Example 8.7

Suppose that in a line of rubber sandals production, we must control the mass of the essential raw material: rubber. Consider that the specification of the rubber mass required for a sandal is (8.0 ± 0.5) g. The scale used to control rubber weighing has an acceptance criterion (error + expanded uncertainty of the scale calibration certificate) less than or equal to 0.1 g and a resolution of 0.01 g. Based on this information, determine the lower and higher values where we have a probability of 95 % acceptance (risk to the 5 % producer).

Consider:

- $T_U = 8.5$ g.
- $T_L = 7.5$ g.
- $u = 0.05$ g ($U/2 = 0.1/2$).[2]

(a) Determining the upper acceptance limit to 95 %

$$Z_U = \left(\frac{T_U - y}{u}\right) \tag{8.9}$$

For 95 %, consulting Table 8.1, we have $z = 1.65$, replacing Eq. 8.9.

$$\left(\frac{8.5 - y}{0.05}\right) = 1.65 \rightarrow y = 8.42\ g$$

(b) Determining the lower acceptance limit to 95 %

$$Z_L = \left(\frac{y - T_L}{u}\right) \tag{8.10}$$

$$1.65 = \left(\frac{y - 7.5}{0.05}\right)$$

$$y = 7.58\ g$$

[2]Consider the coverage factor $k = 2$; 95.45 % of probability.

8.7 Risks Inherent in Decision-Making

In many cases, decision-making may have severe and harmful consequences and may lead to legal, criminal, accusatory, and even health damage.

Next, we will examine how to manage the risks inherent in making decisions based on safe and reliable measurements.

8.7.1 Protected Rejection—Categorical Decision

Protected rejection occurs when the measurement's result will result in a categorical decision, often with legal consequences, causing legal or high risk in decision-making. Let us look at the following example.

Example 8.8 (Source: JGCM 106:2012)
In the application of road law, the road police use devices such as radars to measure car velocity. A decision to issue a speed fine must be made with a high degree of confidence that the speed limit has been exceeded.

Using a radar, speed measurements in the field can be performed with a relative standard uncertainty of 2% within the interval of (50 to 150) km/h. In this interval, the measured velocity v is characterized by a normal distribution with a standard deviation of 0.02 v.

Under these conditions, one may ask: For a limit speed of 100 km/h (the maximum speed at which the driver does not take a fine), what should be the measured velocity by the radar to ensure, with a 99.9 % probability, that the driver exceeded the limit speed of 100 km/h?

This mathematical problem is equivalent to calculating a probability of conformity for a higher unilateral tolerance interval.

Note that z is given by Eq. (8.9) and that the probability of desired trust is 99.9 %, $z = 3.09$ (Table 8.1). Then we have:

$$z = \frac{(T_U - y)}{u}$$

$$3.08 = \frac{(T_U - 100)}{2}$$

$$\mathrm{T_U} = 106.16\ \mathrm{km/h}$$

The interval [100 km/h $\leq v \leq$ 107 km/h] is a guard band that ensures a probability of at least 99.9 %. If the measured speed is over 107 km/h, the police will have 99.98 % confidence that the driver was above 100 km/h, and he will fine the violator.

Table 8.5 TUR = 5, without guard band

m(g)	$\Phi 1\left(\frac{T_U - y}{u}\right)$	$\Phi 2\left(\frac{T_L - y}{u}\right)$	$Pc1$	$Pc2$	Pc
7.50	20.00	0.00	1.00	0.50	50%
7.58	18.40	−1.60	1.00	0.05	95%
7.60	18.00	−2.00	1.00	0.02	98%
7.70	16.00	−4.00	1.00	0.00	100%
7.80	14.00	−6.00	1.00	0.00	100%
7.90	12.00	−8.00	1.00	0.00	100%
8.00	10.00	−10.00	1.00	0.00	100%
8.10	8.00	−12.00	1.00	0.00	100%
8.20	6.00	−14.00	1.00	0.00	100%
8.30	4.00	−16.00	1.00	0.00	100%
8.40	2.00	−18.00	0.98	0.00	98%
8.42	1.60	−18.40	0.95	0.00	95%
8.50	0.00	−20.00	0.50	0.00	50%

8.7.2 Binary Decision Rule Applied to the Conformity Assessment Without Guard Band

Remembering, a binary decision rule exists when the result is limited to two options (approved or disapproved) and a non-binary decision rule when various terms can express the result (approved, conditional approval, conditional disapproval, disapproved).

We will start this theme by analyzing the situation in which we have a binary decision without using the guard band. In a binary statement where the simple acceptance rule is adopted ($w = 0$), we have the situations:

- Approved—the measured value is between the tolerance limits.
- Disapproved—the measured value is outside the tolerance limits.

Let us look at the following example.

Example 8.9
Analyzing the rubber mass specification of Example 8.7, we present the following questions:

1. What should be the uncertainty of the scale that will measure the value of the rubber used in the production of sandals?
2. Should this uncertainty be related to the tolerance of the process?

To answer these two questions, we must adopt a decision rule. Initially, we will adopt a decision rule without a guard band; that is, we are sharing the risk with the consumer of sandals. Knowing that the specification of the mass of rubber used in the production is (8.0 ± 0.5) g, we have $T_L = 7.5$ g and $T_U = 8.5$ g. If we adopt an expanded uncertainty of the scale according to Example 8.7, that is, $U = 0.1$ g (for 95.45 %), we will have (Table 8.5):

Table 8.7 $TUR = 10$. Values for acceptance limit at 95 % equal to 7.54 g and 8.46 g

m(g)	$\Phi 1\left(\frac{T_U - y}{u}\right)$	$\Phi 2\left(\frac{T_L - y}{u}\right)$	$Pc1$	$Pc2$	Pc
7.50	40.00	0.00	1.00	0.50	50%
7.54	38.40	−1.60	1.00	0.05	95%
7.60	36.00	−4.00	1.00	0.00	100%
7.70	32.00	−8.00	1.00	0.00	100%
7.80	28.00	−12.00	1.00	0.00	100%
7.90	24.00	−16.00	1.00	0.00	100%
8.00	20.00	−20.00	1.00	0.00	100%
8.10	16.00	−24.00	1.00	0.00	100%
8.20	12.00	−28.00	1.00	0.00	100%
8.30	8.00	−32.00	1.00	0.00	100%
8.40	4.00	−36.00	1.00	0.00	100%
8.46	1.60	−38.40	0.95	0.00	95%
8.50	0.00	−40.00	0.50	0.00	50%

Note how the guard band reduces and consequently increases the acceptance interval when the *TUR* increases to 10. This increases the chances of product acceptance but dramatically reduces the acceptance criterion and, consequently, the uncertainty of the measuring instrument.

8.8 Binary Decision Rule Applied to the Conformity Assessment with a Guard Band

When we adopt a binary decision rule in the conformity assessment, the measurement result is accepted if the measured value is within the acceptance interval. A value measured outside the acceptance interval leads to the item's rejection.

Using guard bands provides a way to limit the probability of making a decision incorrectly based on measurement information summarized by a probability interval.

The probabilities evaluated depend on two factors: (i) the measurement system and (ii) the production process.

If the measurement system were perfectly accurate, all decision-making would be correct, and the risks would be null. An increase in measurement uncertainty means an increase in the probability of an incorrect decision, and the probability is higher when the measured values are close to the tolerance limits.

Risks also depend on the nature of the production process. If the process rarely produces an item whose properties of interest are close to tolerance limits,[5] there is less opportunity to make incorrect decisions. On the other hand, if a process

[5] Process under control.

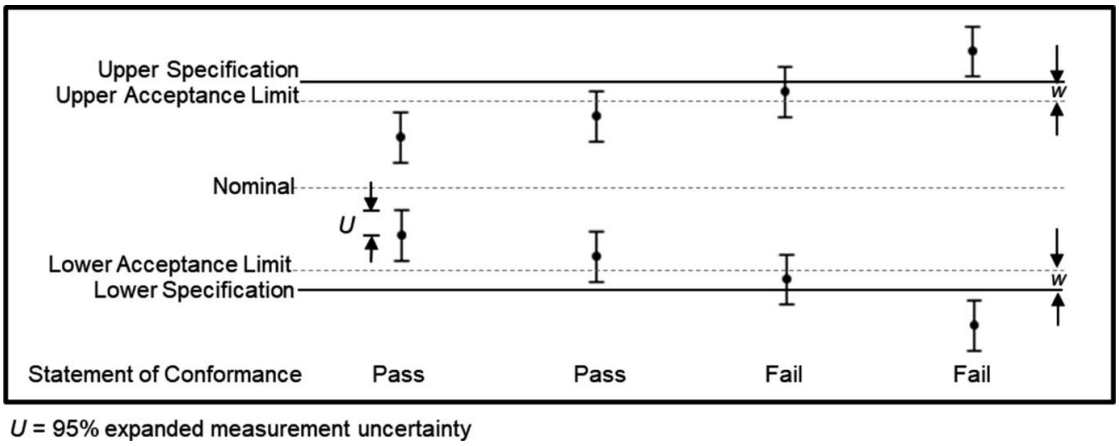


Fig. 8.13 Graphical representation of a binary statement with a guard band. (Source: ILAC G8:09/2019)

produces items with properties that are probably close to tolerance[6] limits, the uncertainties associated with measurements are placed in the game.

An out-of-control process, with a significant dispersion of measured values, produces values near tolerance limits, significantly increasing the probability of wrong decision-making.

The guard band's boundaries for a confidence level of $(1 - \alpha)$ can be obtained, considering a symmetrical error of $\alpha/2$ on each PDF tail.

Statements of conformity are reported as:

- Pass—acceptance is based on the acceptance range; the measurement result is between the acceptance limits.
- Fail—rejection based on the guard band if the measurement result exceeds acceptance limits (Fig. 8.13).

The adoption of a guard band gives protection to risk, according to the size of the band. We adopted the relationship $w = r \cdot U$, where r is the multiplicative factor to the guard band. Let us look at some typical guard bands and their respective *r*.

8.8.1 Guard Band with r = 1 (w = U)—ILAC G8:2009 Rule Decision

In this case, the guard band has the same value as expanded uncertainty. Applying the guard band in the example of sandal production, for a *TUR* equal to five, we have (Table 8.8):

[6] Process out of control.

Table 8.8 $TUR = 5$, guard band $w = U$

m(g)	$\Phi 1\left(\frac{T_U - y}{u}\right)$	$\Phi 2\left(\frac{T_L - y}{u}\right)$	$Pc1$	$Pc2$	Pc
7.50	20.00	0.00	1.00	0.50	50.00%
7.54	19.20	−0.80	1.00	0.21	78.81%
7.60	18.00	−2.00	1.00	0.02	97.72%
7.70	16.00	−4.00	1.00	0.00	100.00%
7.80	14.00	−6.00	1.00	0.00	100.00%
7.90	12.00	−8.00	1.00	0.00	100.00%
8.00	10.00	−10.00	1.00	0.00	100.00%
8.10	8.00	−12.00	1.00	0.00	100.00%
8.20	6.00	−14.00	1.00	0.00	100.00%
8.30	4.00	−16.00	1.00	0.00	100.00%
8.40	2.00	−18.00	0.98	0.00	97.72%
8.46	0.80	−19.20	0.79	0.00	78.81%
8.50	0.00	−20.00	0.50	0.00	50.00%

Adopting $r = 1$, we will always have the acceptance limits with a probability of 97.72 % acceptance at risk of false positives less than 2.5 %.

Thus, for example, at 8.7, we will have a lower acceptance limit equal to 7.60 g and an upper acceptance limit of 8.40 g with a probability of false positive (consumer risk) below 2.5 %.

8.8.2 Guard Band with r = 0.83 (w = 0.83 U)—ISO 14253-1:2017 Rule Decision

In this case, the protection band has a value of 0.83 of expanded uncertainty. We will apply the guard band to the sandal production example for a *TUR* equal to five. With the guard band $w = 0.83\ U$, the risk of false positives is less than 5 % for any value of $TUR > 1$ adopted.

Note that the $r \times U$ relationship is independent of the adopted *TUR* since when we change the *TUR*, we must change the uncertainty of measurement U and, with that, the size of the guard band w.

Thus, for example, at 8.7, we will have a lower acceptance limit equal to 7.58 g and an upper acceptance limit of 8.42 g, with a probability of false positive (consumer risk) below 5 %. The guard band will be equal to $w = 0.83 \times 0.1 = 0.083$ g (Table 8.9).

$$A_U = 8.50–0.083 = 8.471 = 8.42\ \text{g}$$

$$A_L = 7.50 + 0.083 = 7.583 = 7.58\ \text{g}$$

Table 8.9 TUR = 5, guard band $w = 0.83\ U$

m(g)	$\Phi 1\left(\frac{T_U - y}{u}\right)$	$\Phi 2\left(\frac{T_L - y}{u}\right)$	$Pc1$	$Pc2$	Pc
7.50	20.00	0.00	1.00	0.50	50.00%
7.58	18.34	−1.66	1.00	0.05	95.15%
7.60	18.00	−2.00	1.00	0.02	97.72%
7.70	16.00	−4.00	1.00	0.00	100.00%
7.80	14.00	−6.00	1.00	0.00	100.00%
7.90	12.00	−8.00	1.00	0.00	100.00%
8.00	10.00	−10.00	1.00	0.00	100.00%
8.10	8.00	−12.00	1.00	0.00	100.00%
8.20	6.00	−14.00	1.00	0.00	100.00%
8.30	4.00	−16.00	1.00	0.00	100.00%
8.40	2.00	−18.00	0.98	0.00	97.72%
8.42	1.66	−18.34	0.95	0.00	95.15%
8.50	0.00	−20.00	0.50	0.00	50.00%

But if we consider, for example, a *TUR* equal to two, we will have $U = 0.25$ and $w = 0.25 \times 0.83 = 0.23$.

$$A_U = 8.50 - 0.23 = 8.27\ \text{g}$$

$$A_L = 7.50 + 0.23 = 7.73\ \text{g}$$

With the reduction of *TUR*, measurement uncertainty increases. This generates a lower acceptance interval, which may imply more rejected measurements.

8.8.3 Guard Band with r = 1.5 (w = 1.5 U)—Three Sigma Rule Decision

In this case, the protection band is 1.5 times expanded uncertainty. Applying the guard band in the example of sandal production, for a *TUR* equal to five, we have (Table 8.10):

With the guard band $w = 1.5$ U, we have, for any value of $TUR > 1$ adopted, a risk of false positive less than 0.16 % (100 % − 99.87 % = 0.13 %).

Thus, for example, at 8.7, we will have a lower acceptance limit equal to 7.65 g and an upper acceptance limit of 8.35 g, with a probability of false positives (consumer risk) below 0.16 %. The guard band will be equal to $w = 1.5 \times 0.1 = 0.15$ g.

Table 8.10 $TUR = 5$, guard band $w = 1.5\ U$

m(g)	$\Phi 1\left(\frac{T_U - y}{u}\right)$	$\Phi 2\left(\frac{T_L - y}{u}\right)$	$Pc1$	$Pc2$	Pc
7.50	20.00	0.00	1.00	0.50	50.00%
7.60	18.00	−2.00	1.00	0.02	97.72%
7.65	17.00	−3.00	1.00	0.00	99.87%
7.70	16.00	−4.00	1.00	0.00	100.00%
7.80	14.00	−6.00	1.00	0.00	100.00%
7.90	12.00	−8.00	1.00	0.00	100.00%
8.00	10.00	−10.00	1.00	0.00	100.00%
8.10	8.00	−12.00	1.00	0.00	100.00%
8.20	6.00	−14.00	1.00	0.00	100.00%
8.30	4.00	−16.00	1.00	0.00	100.00%
8.35	3.00	−17.00	1.00	0.00	99.87%
8.40	2.00	−18.00	0.98	0.00	97.72%
8.50	0.00	−20.00	0.50	0.00	50.00%

Table 8.11 Probability of false positive (FP) for any $TUR > 1$

Decision rule	Guard band w	Risk
Six Sigma	3 U	<1 ppm FP
Three Sigma	1.5 U	<0.16 % FP
ILAC G8:2009	U	<2.5 % FP
ISO 14253-1:2017	0.83 U	<5 % FP
Simple acceptance	0	<50 % FP
Defined by the customer	$r \times U$	Customers can define r multiple arbitraries to apply as a guard band

8.8.4 *Guard Band with r = 3 (w = 3 U)—Six Sigma Rule Decision*

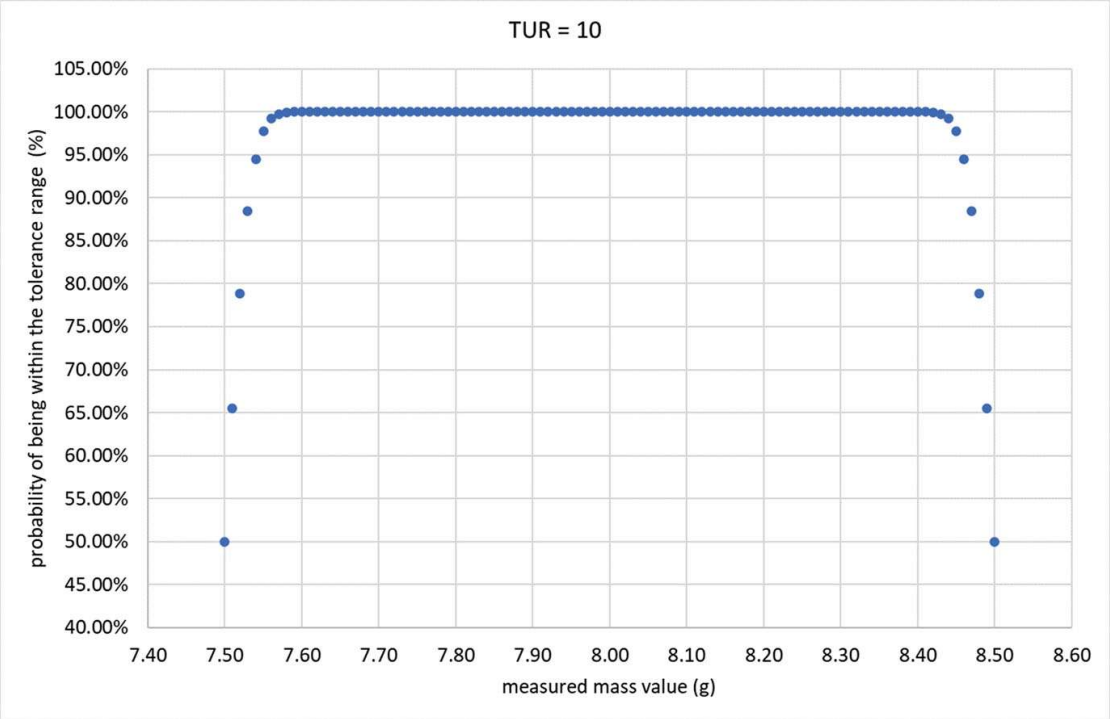
In this case, the protection band is three times expanded uncertainty. With the guard band $w = 3\ U$, we have, for any value of $TUR > 1$ adopted, a risk of false positive less than 1 ppm (0.0001 %).

Thus, at 8.7, we will have a lower acceptance limit equal to 7.80 g (7.50 + 0.3) and an upper acceptance limit of 8.20 g (8.50 − 0.3).

Table 8.11, copied from ILAC G8:09/2019 Guidelines on Decision Rules and Statements of Conformity, summarizes some of the guard band values presented in this chapter.

As we can observe in Graphs 8.1, 8.2, 8.3, 8.4 and 8.5, mass values between the tolerance limits increase with the highest probability when the *TUR* increases. This entails greater security in the results obtained.

Note that when *TUR* equals 1, a few values are approved for a given probability, for example, 95 %. For this reason, we should not adopt *TUR* equal to 1. When *TUR* equals 2, the values with the highest probability of occurrence within tolerance limits



Graph 8.5 Probability × measured mass, for example 8.7 with *TUR* equal to 10

8.9 Proposed Exercises

8.9.1 A "specific risk" is a probability that an accepted item is not in compliance or a rejected item is in accordance. Considering this definition, tick the correct alternative.

(a) The "specific consumer risk" is the probability of accepting an item according to, and the "specific risk of the producer" is the probability of rejecting an item not according to.
(b) The "specific consumer risk" is the probability of accepting an item not according to, and the "specific risk of the producer" is the probability of rejecting an item according to.
(c) The "specific risk of the producer" is the probability of accepting an item not according to, and the "specific consumer risk" is the probability of rejecting an item according to.
(d) The "specific consumer risk" and the "specific risk of the producer" will always be the same.

8.9.2. Check the alternative that best defines a binary decision rule.

(a) The result is limited to two options: approved/disapproved.
(b) We have two options; however, a guard band will always exist.
(c) When there is no possibility of using a non-binary decision rule.